Exp. No.: 1	Study of Various Network Commands Used in Linux and Windows

Aim

To study various network commands used in Linux and Windows operating systems and to gain hands-on practice in using these commands for network configuration, diagnosis, and troubleshooting.

Theory / Concepts

Windows Networking Commands

`arp -a`

ARP (Address Resolution Protocol) command shows the IP address of the computer along with the IP address and MAC address of the router.

`hostname`

The simplest of all TCP/IP commands; it displays the name of the computer.

`ipconfig /all`

Displays detailed configuration information about the TCP/IP connection, including Router, Gateway, DNS, DHCP, and the type of Ethernet adapter in the system.

`nbtstat -a`

Helps solve problems with NetBIOS name resolution (Nbt stands for NetBIOS over TCP/IP).

`netstat`

Displays a variety of statistics about a computer's active TCP/IP connections. It is used for monitoring incoming and outgoing network connections, routing tables and interface statistics (e.g., netstat -r).

`nslookup`

A tool used to perform DNS lookups, such as finding the IP address of a computer, the MX records for a domain, or the NS servers of a domain. It can operate in interactive and non-interactive modes (e.g., nslookup www.google.com).

`pathping`

Unique to Windows; a combination of the Ping and Tracert commands. It traces the route to the destination address and then runs a 25-second test of each router along the way, gathering statistics on the rate of data loss at each hop.

`ping`

Packet INternet Groper — the best way to test connectivity between two nodes. Ping uses ICMP (Internet Control Message Protocol) to communicate with other devices, e.g. ping hostname, ping IP address, ping fully qualified domain name.

`route`

Used to show or manipulate the IP routing table; primarily used to set up static routes to specific hosts or networks via an interface.

Important Linux Networking Commands

1. ip

The ip command is one of the basic commands every administrator needs for daily work, from setting up new systems and assigning IPs to troubleshooting existing systems. It can show address information, manipulate routing, and display network devices, interfaces and tunnels. General syntax: ip <OPTIONS> <OBJECT> <COMMAND>

```
# Show IP addresses assigned to an interface
ip address show
```

```
# Assign an IP address to an interface (e.g., enp0s3)
ip address add 192.168.1.254/24 dev enp0s3

# Delete an IP address from an interface
ip address del 192.168.1.254/24 dev enp0s3

# Bring an interface online / offline
ip link set eth0 up
ip link set eth0 down

# Enable promiscuous mode on an interface
ip link set eth0 promisc on

# Add a default route via a gateway
ip route add default via 192.168.1.254 dev eth0

# Add / delete a route to a network
ip route add 192.168.1.0/24 via 192.168.1.254
ip route add 192.168.1.0/24 dev eth0
ip route delete 192.168.1.0/24 via 192.168.1.254

# Display the route taken for a given IP
ip route get 10.10.1.4
```

2. ifconfig

The ifconfig command was a staple tool for configuring and troubleshooting networks. It has since been replaced by the ip command described above.

3. mtr (Matt's Traceroute)

A command-line network diagnostic tool that combines the functionality of ping and traceroute. It shows the route from the computer to a specified host along with statistics for each hop, such as response time and packet loss percentage. Syntax: mtr <options> hostname/IP

```
mtr google.com          # basic statistics for each hop
mtr -g google.com       # show numeric IP addresses
mtr -b google.com       # show IP addresses and hostnames
mtr -c 10 google.com    # set the number of pings to send
```

4. tcpdump

Designed for capturing and displaying network packets.

```
dnf install -y tcpdump      # install tcpdump
tcpdump -D                  # list available capture interfaces
tcpdump -i eth0              # capture traffic on eth0
tcpdump -i eth0 -c 10        # capture 10 packets on eth0

# Filter traffic to/from a specific host
tcpdump -i eth0 -c 10 host 8.8.8.8
tcpdump -i eth0 src host 8.8.8.8
tcpdump -i eth0 dst host 8.8.8.8

# Filter traffic to/from a network
tcpdump -i eth0 net 10.1.0.0 mask 255.255.255.0
tcpdump -i eth0 net 10.1.0.0/24

# Filter traffic by port
tcpdump -i eth0 port 53
tcpdump -i eth0 host 8.8.8.8 and port 53
```

```
tcpdump -i eth0 -c 10 host www.google.com and port 443
tcpdump -i eth0 port not 53 and not 25
```

5. ping

Verifies IP-level connectivity to another TCP/IP computer by sending ICMP Echo Request messages; the corresponding Echo Reply messages are displayed along with round-trip times. Ping is the primary command used to troubleshoot connectivity, reachability and name resolution.

```
ping google.com          # ping continuously (Ctrl+C to stop)
ping -c 10 google.com    # ping a host 10 times
```

Common ping results include:

- Destination Host Unreachable — there is no route from the local host to the destination, or a remote router reports it has no route to the destination host.
- Request timed out — no Echo Reply was received within the default time; possible causes include network congestion, ARP request failure, or packet filtering/firewall rules.
- Unknown host / Ping Request Could Not Find Host — the hostname may be misspelled or the host does not exist on the network.

A good ping result shows 0% packet loss with low latency; the exact latency depends on the transmission medium used (UTP, fibre optic cable, or Wi-Fi).

Procedure

Configuring an Ethernet connection using nmcli

If a host is connected to the network over Ethernet, its connection settings can be managed from the command line using the nmcli utility.

1. List the NetworkManager connection profiles:

```
nmcli connection show
```

2. Add a new connection profile (skip this step to modify an existing profile):

```
nmcli connection add con-name <connection-name> ifname <device-name> type ethernet
```

3. Optional: rename the connection profile:

```
nmcli connection modify "Wired connection 1"
```

4. Display the current settings of the connection profile:

```
nmcli connection show
```

5. Configure the IPv4 settings — to use DHCP:

```
nmcli connection modify "Wired connection 1" ipv4.method auto
```

To set a static IPv4 address, network mask, default gateway, DNS servers and search domain:

```
nmcli connection modify "Wired connection 1" ipv4.method manual \
  ipv4.addresses 192.0.2.1/24 ipv4.gateway 192.0.2.254 \
  ipv4.dns 192.0.2.200 ipv4.dns-search example.com
```

6. Configure the IPv6 settings — to use stateless address autoconfiguration (SLAAC):

```
nmcli connection modify "Wired connection 1" ipv6.method auto
```

To set a static IPv6 address, network mask, default gateway, DNS servers and search domain:

```
nmcli connection modify "Wired connection 1" ipv6.method manual \
  ipv6.addresses 2001:db8:1::fffe/64 ipv6.gateway 2001:db8:1::fffe \
  ipv6.dns 2001:db8:1::ffbb ipv6.dns-search example.com
```

7. Activate the profile:

```
nmcli connection up Internal-LAN
```

Verification

1. Display the IP settings of the network interface card:

```
ip address show enp1s0
```

2. Display the IPv4 default gateway:

```
ip route show default
```

3. Display the IPv6 default gateway:

```
ip -6 route show default
```

4. Display the DNS settings:

```
cat /etc/resolv.conf
```

5. Verify that the host can send packets to other hosts:

```
ping <host-name-or-IP-address>
```

Troubleshooting

- Verify that the network cable is plugged into the host and the switch.
- Check whether the link failure exists only on this host or also on other hosts connected to the same switch.
- Verify that the network cable and the network interface are working as expected; perform hardware diagnosis and replace defective cables or network interface cards.
- If the configuration on disk does not match the configuration on the device, starting or restarting NetworkManager creates an in-memory connection reflecting the device's current configuration.

Output

Sample output of the ping command:

```
$ ping google.com
PING google.com (216.58.206.174) 56(84) bytes of data.
64 bytes from sof02s27-in-f14.1e100.net (216.58.206.174): icmp_seq=1 ttl=56 time=10.7
ms
64 bytes from sof02s27-in-f14.1e100.net (216.58.206.174): icmp_seq=2 ttl=56 time=10.2
ms
64 bytes from sof02s27-in-f14.1e100.net (216.58.206.174): icmp_seq=3 ttl=56 time=10.4
ms
^C
```

Result

Various network commands used in Linux and Windows for configuration, diagnosis, and troubleshooting were studied, and hands-on practice was carried out successfully.